Customer Churn & Revenue Intelligence Platform

Tableau Dashboard Documentation | Nehali Parulekar

Data Analyst | Last Updated: June 2026

1. Purpose and Business Context

This dashboard was built to answer a core business question: which customers are likely to leave, how much revenue does that represent, and where should retention efforts be focused?

It is designed for two audiences: business stakeholders who need clear KPIs and actionable retention signals, and technical reviewers who want to see how the data was structured and analyzed. The dashboard walks through churn behavior, revenue exposure, customer segmentation, service usage patterns, and a prioritized list of at-risk customers.

2. At a Glance: Key Metrics

Metric	Value
Total Customers	7,043
Overall Churn Rate	26.5%
Monthly Recurring Revenue (MRR)	$456.1K
MRR at Risk (from churn)	$139.1K
Average Customer Lifetime Value	$2,279.6
Average Tenure	32.4 months
High-Risk Customers	970
Recoverable MRR (Priority 1 + 2)	$47.8K / month

3. Data Source and Pipeline

Dataset: IBM Telco Customer Churn (publicly available via Kaggle / IBM Sample Data)

The dataset contains 7,043 customer records with attributes covering contract type, internet service, payment method, monthly charges, tenure, add-on service subscriptions, and demographics.

Pipeline Architecture

- **raw schema:** Raw CSV loaded into PostgreSQL without modification (audit trail layer)
- **staging schema:** Type casting, null handling, and column normalization applied here
- **dw schema:** Star schema with four dimension tables and one fact table -- query layer for Tableau

Data Quality Notes

- TotalCharges stored as text with 11 blank rows (tenure = 0 customers). Converted to NUMERIC; blanks set to NULL, not zero, to avoid corrupting revenue aggregations.

- SeniorCitizen stored as 0/1 integer while all other binary fields use Yes/No. Normalized to Yes/No in staging.
- No duplicate customer records. Dataset is already deduplicated at the customer level.

4. Color Coding Logic

The dashboard uses a consistent color system across all tabs:

Color	Meaning
Red	Churned customers, high churn rates, MRR at risk
Amber	High risk segments / Retained customers, low churn rates, stable segments
Yellow	Medium risk segments and secondary comparisons
Blue	Low risk or no-service segments

5. Dashboard Structure: Tab-by-Tab Guide

Tab 1 -- Overview

High-level summary designed for executives and stakeholders who need the headline numbers without drilling into detail.

- Total customers, churn rate, MRR, MRR at risk, avg CLV, avg tenure (KPI cards)
- Churn by Contract Type (horizontal bar chart) -- month-to-month at 42.7%, two-year at 2.8%
- MRR at Risk Breakdown (donut chart) -- month-to-month contracts account for $120.8K of the $139.1K total

Tab 2 -- Churn Analysis

Detailed churn breakdown. Filter by dependents, partner status, senior citizen flag, and tenure bucket to isolate specific customer groups.

- Churn Rate by Contract Type (bar with avg reference line)
- Churn Rate by Internet Service (bar with avg reference line)
- Churn Rate by Payment Method (bar with avg reference line)
- Churn Rate by Tenure Bucket (bar + trend line) -- 0-12 months is the highest-risk window
- Churn by Contract vs Internet Service (cross-tab highlight table) -- month-to-month + fiber optic: 57.9%

Tab 3 -- Revenue Analysis

Revenue exposure and distribution. Filterable by contract type and internet tier.

- MRR by Contract Type (stacked bar showing retained vs churned revenue)
- Avg CLV by Contract Type (bar) -- two-year customers have the highest estimated lifetime value
- MRR by Internet Tier (bar) -- fiber optic generates the most monthly revenue
- Monthly Charges Distribution (histogram with churn overlay)
- Revenue Concentration (scatter) -- identifies high-charge outlier customers

Tab 4 -- Customer Segmentation

Segments customers into four risk tiers. Filterable by senior segment, gender, and contract type.

- Risk Segment Summary -- Churned: 26.5%, High Risk: 13.8%, Medium Risk: 17.8%, Low Risk: 41.9%
- Household Type vs Churn Rate -- single customers without dependents churn at the highest rate
- Senior vs Non-Senior churn comparison
- Churn by Add-on Count -- clear inverse relationship (0 add-ons: 37.7%, 6+ add-ons: 5.3%)
- Segment Revenue Table -- MRR, avg MRR, and customer count by segment and contract type

Tab 5 -- Services

Compares churn rates between subscribers and non-subscribers across all 8 service types. Filterable by contract type and internet service.

- Churn Rate by Service: Subscribers vs Non-Subscribers -- subscribers churn at lower rates for every service type
- Service Adoption Rate (% of customers enrolled per service)
- Internet Service Breakdown (customer count by tier)
- Avg Monthly Charges by Internet Service + Contract Type (grouped bar)

Tab 6 -- Explorer & Retention

Actionable retention view. Pre-filtered to high-risk customers on month-to-month contracts. Designed for a customer success or retention team to use directly.

- KPI summary: 970 high-risk customers, 125 Priority 1, $47.8K MRR at risk, avg 5 month tenure
- Priority 1 vs Priority 2 breakdown (ranked by monthly charge value)
- Customer Detail Table -- customer ID, risk segment, recommendation, churn flag, contract, internet service, payment method, monthly and estimated charges
- Estimated Monthly Revenue Recovery chart -- projected MRR recovered if high-risk customers are retained

6. Key Findings

Contract type is the strongest churn predictor: Month-to-month customers churn at 42.7% vs 11.3% for one-year and 2.8% for two-year contracts. Month-to-month contracts represent $120,847 of the $139.1K total MRR at risk.

Fiber optic drives churn despite being the highest-revenue service tier: Fiber optic customers churn at nearly 30%, almost double the DSL rate. The combination of month-to-month contract and fiber optic service produces a 57.9% churn rate -- the highest in the dataset.

Payment method is a signal, not just a preference: Electronic check payers churn at 45.3%. Customers on automatic payment methods (bank transfer or credit card auto-pay) churn at roughly 15%. This may indicate engagement and commitment to the service.

Add-on services are the clearest retention lever: Churn drops from 37.7% at 0 add-ons to 5.3% at 6+ add-ons. Subscribers churn at lower rates than non-subscribers across all 8 service types without exception. Encouraging service adoption reduces churn.

Early tenure is the highest-risk window: Customers in the 0-12 month bucket have the highest churn rate. Churn decreases steadily beyond 12 months and stabilizes significantly after 37 months. Early engagement and onboarding are the highest-value retention investments.

970 high-risk customers represent a concrete retention opportunity: These customers are on month-to-month contracts, have not churned yet, and are classified as high risk based on contract type, service profile, and tenure. Retaining them is worth an estimated $47.8K/month.

7. Technical Build Details

Component	Detail
Database	PostgreSQL (local) -- three-schema design: raw, staging, dw
Data Modeling	Star schema: dim_customer, dim_contract, dim_service_plan, dim_date, fact_subscription
Data Cleaning	SQL staging layer: type casting, null handling, normalization
Data Preparation	Python (Pandas, NumPy) -- data inspection, loading, validation, EDA
EDA Charts	Python (Matplotlib, Seaborn, SciPy) -- 7 charts covering distributions, correlations, outliers
Visualization	Tableau Desktop -- 6-tab story with cross-tab, bar, scatter, donut, KPI cards
Publishing	Tableau Public
Dataset	IBM Telco Customer Churn (public, via Kaggle)

Calculated Fields in Tableau

- **Churn Rate:** COUNT(churned customers) / COUNT(all customers)
- **MRR at Risk:** SUM of monthly charges for churned customers by segment
- **Estimated CLV:** AVG(monthly charges) x AVG(tenure months) -- projected, not modeled
- **Risk Segment:** Rule-based classification using churn flag, contract type, and tenure bucket
- **Priority Flag:** Derived field ranking high-risk customers by monthly charge value (descending)

8. Limitations and Assumptions

- This is a publicly available sample dataset. It does not represent any real company or actual customer behavior.
- CLV estimates are projections based on average monthly charges multiplied by average tenure. They are not modeled lifetime values.
- The risk segmentation logic is rule-based, not a predictive machine learning model.
- Revenue recovery estimates assume 100% retention success for the high-risk group, for illustration purposes only.
- Priority 1 vs Priority 2 is determined by monthly charge value (descending). It is not based on propensity modeling.

9. Project Links and Contact

Tableau Public	http://public.tableau.com/app/profile/nehalip
GitHub Repository	https://github.com/parulekarnehali/customer-churn-revenue-intelligence
LinkedIn	https://www.linkedin.com/in/nehalip/
Email	nehaliparulekar0395@gmail.com